Python Programming

Subject: Computer Science | NoteShare Library

PYTHON PROGRAMMING

1. BASICS
Variables: no type declaration
print("Hello")
input() for reading

2. DATA TYPES
int, float, str, bool
list, tuple, set, dict
type(x), isinstance(x, int)

3. STRINGS
s.upper(), s.lower()
s.split(), s.strip()
s.replace()
f-strings: f"Hi {name}"

4. CONTROL
if/elif/else
for x in range(5)
while cond

5. FUNCTIONS
def f(a, b=0): return a+b
*args, **kwargs
lambda x: x**2

6. LISTS
append, insert, remove, pop
sort, reverse
Slicing: [1:3], [::-1]
Comprehensions: [x**2 for x in range(10)]

7. DICTS
person = {"name": "John"}
person["age"] = 25
.get(key, default)

8. OOP
class Animal:
def __init__(self, name):

self.name = name
def speak(self):
return "..."
class Dog(Animal):
def speak(self):
return "Woof"

9. FILE I/O
with open("f.txt", "r") as f:
content = f.read()

10. ERROR HANDLING
try/except/finally
ValueError, ZeroDivisionError

11. LIBRARIES
requests, numpy, pandas
matplotlib, flask, django